

Novel and Simple Cross-sectional FIB Circuit Edit Techniques for Circuit Isolation

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Abstract— Focused Ion Beam Circuit Edit (FIB-CE) is one of the most commonly used silicon (Si) debug tools that are often used in the semiconductor industry, especially for the new design prototype chip. With the expanding growth of semiconductor applications in different areas, such as next-generation smartphones, automotive, medical equipment, artificial intelligence, RF application, and high power devices, the new prototype chip demand has also increased. Hence, FIB-CE plays an important role in Si debugging for new designs or prototype chips. Among FIB-CE jobs in failure analysis (FA), circuit isolation is one of the common requests. This paper introduces a novel and simple cross-sectional FIB circuit edit (XFIB-CE) techniques for circuit isolation. This technique utilized a normal cross-sectional FIB as a method to cut and isolate the unwanted circuitry. Hence, it is an easier, simpler and more efficient alternative technique for FIB-CE circuit isolation as compared with the top-down FIB-CE milling technique.

Keywords— FIB-CE, Circuit Edit, cross-sectional FIB, circuit isolation, XFIB-CE (key words)

I. INTRODUCTION

The semiconductor industry, the unsung heroes of all modern-era digital devices and future technologies, is expanding its scope in next-generation smartphones, automotive, medical equipment, artificial intelligence, radio frequency (RF) applications, high power devices, etc. They are fast becoming a necessary technology for nearly every electronic device. It has registered multi-fold growth in the past decade in terms of revenue generation. With the expanding growth of its application, the new prototype chip demand also increases.

New design chips or prototypes of next-generation chips often face design issues. To understand or confirm the design issue, electrical fault isolation (EFI), and Focused Ion Beam (FIB) Circuit Editing (FIB-CE) are one of the common techniques that use for Silicon (Si) debugging [1-2]. One of the most commonly used techniques in FIB-CE is to perform top-down milling to isolate the suspected circuitry and retest it on a circuit board [1-3]. This paper introduces and demonstrates circuit isolation with a novel and simple cross-sectional FIB technique instead of the conventional layer-by-layer top-down FIB milling method. The proposed technique not only decreases the complexity of sample preparation but also reduced sample preparation time, especially during batch preparation of samples. Besides this, the proposed technique can be applied to FIB-CE on a packaged sample at a location near to wire-bonded location without damaging the bonded gold wires. Conventional top-down FIB-CE required a few gas injection systems (GIS) to perform the milling on different materials such as Aluminium (Al), Copper (Cu), and dielectric layers in the chip so that to mill away or isolate the target circuitry. However, during the insertion of the GIS nozzle, the nozzle tip will damage the bonded wire if the

target location is too near to the bonded gold wire ball. The proposed new technique can easily resolve this issue.

II. EXPERIMENT AND METHODOLOGY

Similar to the conventional FIB-CE, the circuitry (GDS) layout assessment is required to study the feasibility. Follow by developing the FIB-CE recipe on dummy samples before working on the real sample. Figure 1 shows the typical FIB-CE flow, the proposed technique also follows the same flow, the only differences from the conventional method are the layout study feasibility and recipe development. The proposed technique makes use of a normal cross-sectional FIB (XFIB) surface as an isolation technique. Figure 2 illustrates the new proposed XFIB circuit edit (XFIB-CE) concept to isolate a metal line from other connected circuitry. An example of metal3 (M3) that connected other circuitry needs to be cut and isolate from the main circuitry. In the layout study, we will find the connection area without any important net-link or metal line surrounding it to avoid 1) cutting other circuitry, 2) FIB re-deposition (re-dep) leakage problems, 3) induced artifacts, etc. This technique makes use of XFIB to cut/break the connected metal to isolate the unwanted circuitry as shown in Figure 2. However, to avoid over the cut to active for certain cases, recipe development or experiment to monitor the cut was required. This part includes assessing the layout to make sure the spacing of the interest (x) is equal to the interest metal depth (z) we need to mill (Fig. 2). Although the angle of our FIB system is 52° tilted from the plane surface, to have better tolerance for the FIB cut, we use 45° , so that the cross-section surface to the unwanted circuitry spacing x is equal to the depth of cut metal, z .

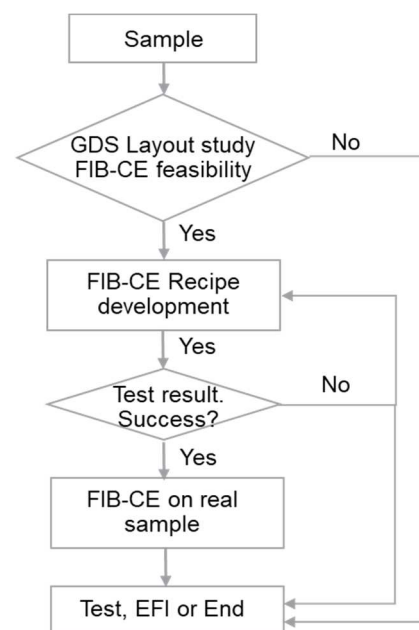


Fig. 1. A standard FIB-CE flow chart.

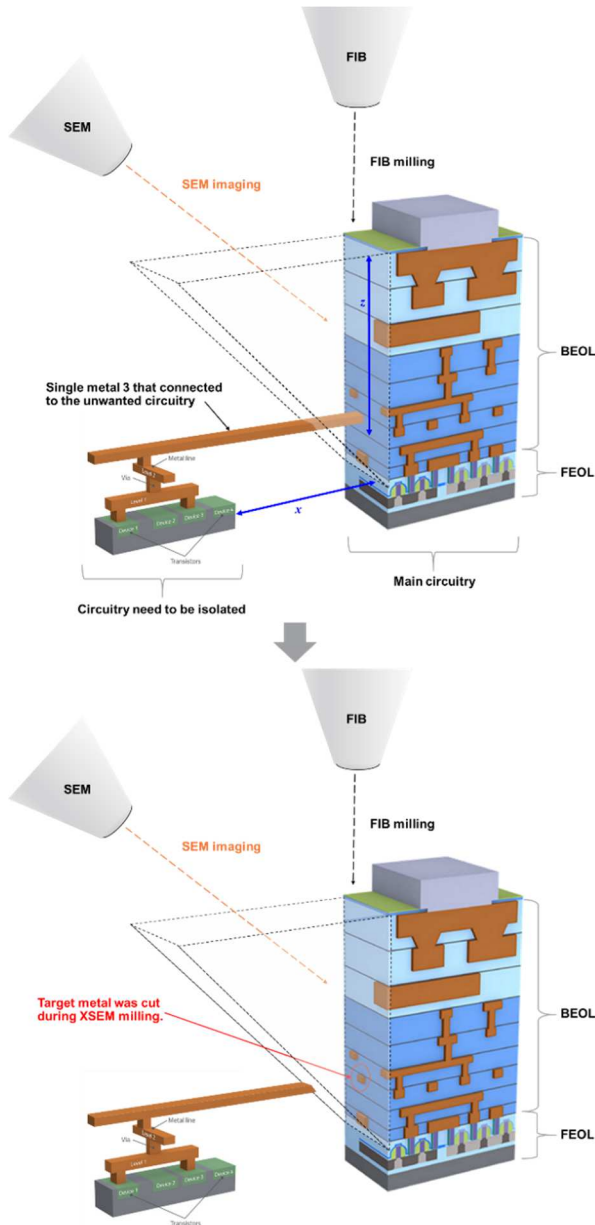


Fig. 2. Concept of the proposed cross-sectional FIB (XFIB) surface isolation technique, the top image shows before XFIB, and the bottom image shows after XFIB.

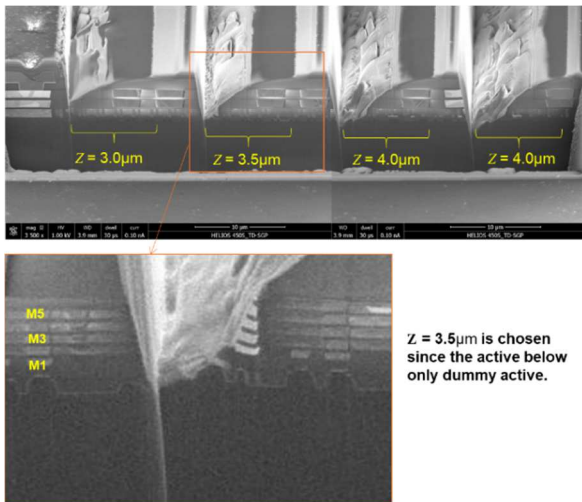


Fig. 3. Cross-sectional images of different milling z depth settings in FIB.

The FIB system used in the experiment is dual-beam FIB Helios 450S from FEI, which is a commercial FIB that is used to prepare Transmission Electron Microscope (TEM) lamella and progressive XFIB for defect identification [4-6]. To experiment with the milling depth, a different z depth setting in FIB were used, Figure 3 show the cross-sectional of different z depth milling setting. For this example, the targeted layer is M3, the best setting will be $z = 3.5\mu\text{m}$; $z = 3.0\mu\text{m}$ is under milled, milling depth only reaches Metal6 (M6), where $z \geq 4.0\mu\text{m}$ is over milled until substrate area, which potentially caused leakage if got a re-dep issue. After confirming the best condition, then we will work on the actual sample. An insulator layer will deposit on the XFIB-CE surface to prevent Cu metal migration or corroding after being exposed to the ambient environment. Finally, the XFIB-CE sample/package will be sent for testing.

III. CASE STUDIES

In this section, three case studies from different technology nodes, wafer level, and package level for XFIB-CE will be discussed.

A. Case study 1

The first case study is on a 55nm tech node 10V PMOS (P-channel Metal-Oxide Semiconductor) BCDlite (Bipolar-CMOS-DMOS) process Bipolar Junction Transistor (BJT) having problems measuring an electrical test (ET) parasitic BJT value. The process integration (PI) and design team suspect the plasma-induced damaged (PID) protection circuit connected parallel to the BJT affects the measurement. Figure 4 shows the ET GDS layout of the 500 fingers PMOS BJT with a PID protection circuit. PI team requested to isolate the PID circuit by FIB-CE, the PID structure is connected at Metal2 (M2), and on top of M2 got many dummy metals (Fig. 4). Since this wafer got 6 Cu metal layers + 1 Aluminum (Al) pad layer, top-down FIB-CE will be very time-consuming (Fig. 5). PI requests at least 5 sites of 500 fingers BJT ET to be measured once the first FIB-CE sample is successful, in total 5 FIB-CE cuts were required. To reduce the sample preparation time, we proposed the XFIB-CE technique, and use normal cross-sectional FIB to cut and isolate the M2.

Figure 6 shows the optical microscope (OM) images and scanning electron microscope (SEM) images before and after XFIB-CE on the PID of 500 fingers BJT ET structure. After fine milling the cross-sectional of M2, we deposit an insulator layer with GIS. Figure 7 shows the parasitic measurement results before and after XFIB-CE, the results meeting their expectation. With this simple and easy XFIB-CE, the sample preparation time can be reduced at least 8x than normal top-down FIB-CE, a very efficient and simple technique.

B. Case study 2

Case study 2 is an $0.13\mu\text{m}$ tech node analog device having high leakage current issue. The design team found the analog cell (SIG_5V) is connected to a 1.5V thin gate input device, therefore the 5V voltage sweeping is limited by the thin gate input circuitry resulting in high leakage. The measurement result does not represent the analog cell leakage value. The design team requested to FIB-CE cut a M3 to isolate the analog cell from the thin gate input device.

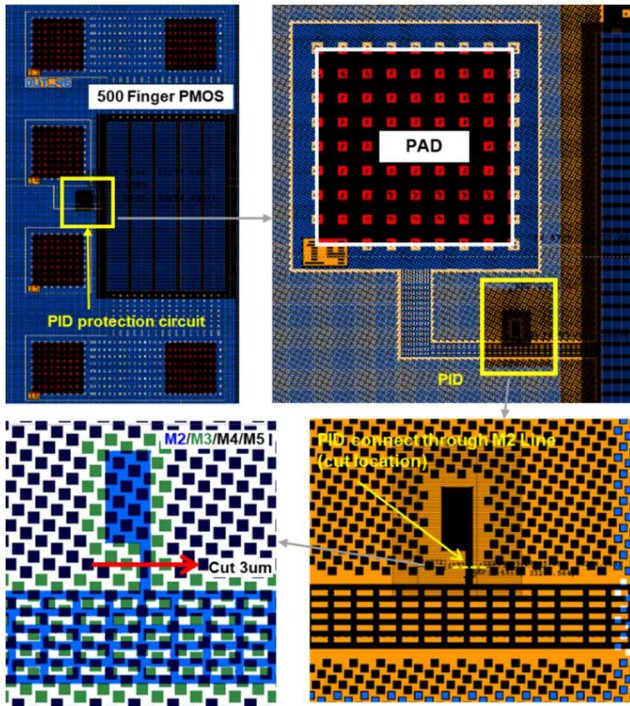


Fig. 4. GDS layout of 500 finger PMOS BJT ET with a PID protection circuit connected parallel at metal2 (M2) layer.

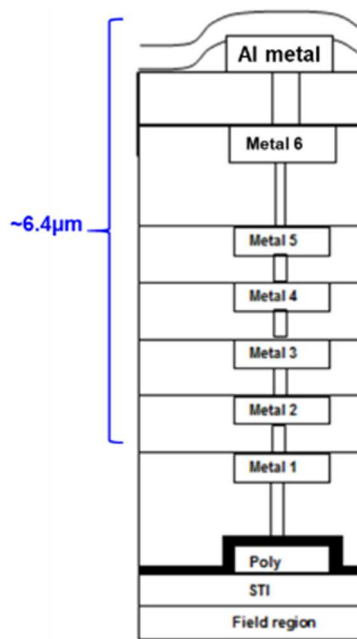


Fig. 5. Cross-sectional illustration of the BCDlite process sample, the depth of M2 $\sim 6.4\mu\text{m}$ from layer process surface.

Figure 8 shows the GDS layout of the M3 (Green line in red box) that is connected to the thin gate input device that needs to be cut. To prevent the leakage that is potentially created at the FIB milling trench sidewall (SEM images shown in Fig. 6), we proposed to XFIB-CE at 45° (purple dotted line box in Fig. 8) instead of XFIB-CE vertically or horizontally (blue dotted line box in Fig. 8). This not only prevents the re-dep artifacts but also provides bigger spacing for the trench opening. The sample for this case is in lead-frame package form, and it needs to be tested in the socket after circuit editing.

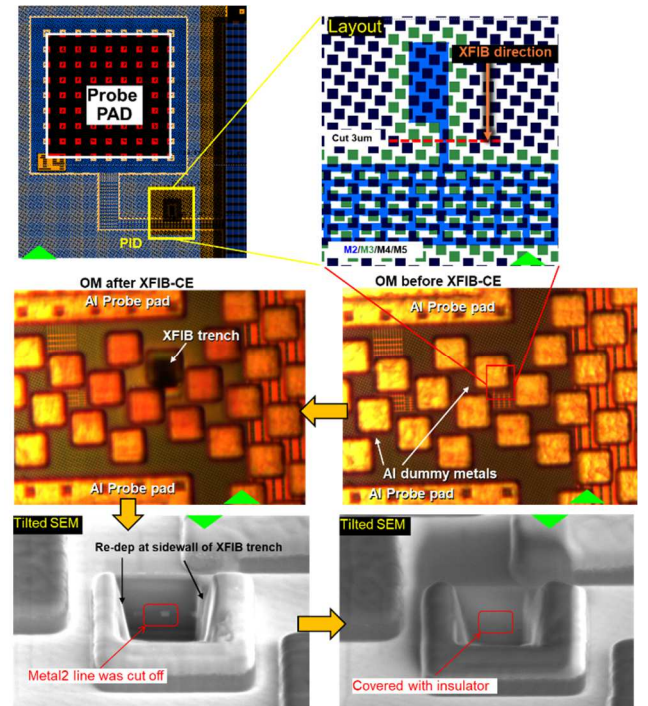


Fig. 6. OM and SEM images of before and after XFIB-CE, the targeted PID connected M2 was cut and isolated.

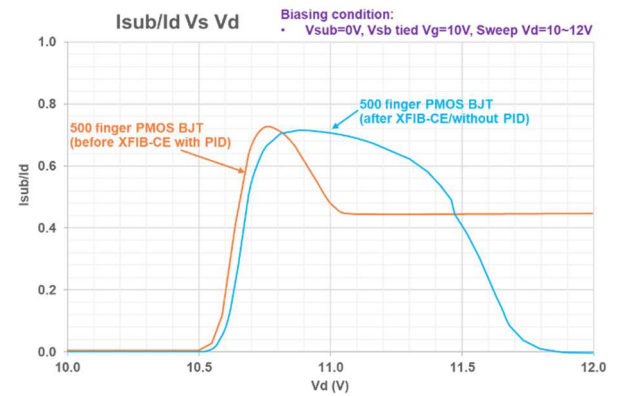


Fig. 7. Current ratio-Voltage (IV) measurement results of the sample before and after XFIB-CE, the after XFIB-CE results meeting PI's expectation.

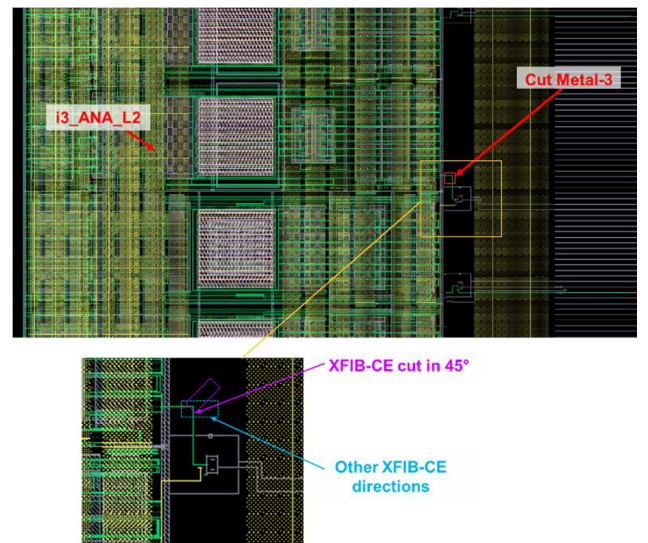


Fig. 8. GDS layout of the M3 (Green colour line in red box) that needs to be cut to isolate the thin gate device. A purple dotted line shows the chosen XFIB-CE location.

Figure 9 shows the lead-frame package, and after partial decapped photos for XFIB-CE. The targeted M3 depth for this sample is $\sim 6.4\mu\text{m}$ from the sample surface. Similar to case study 1, 1 sample was used for recipe development followed by XFIB-CE on a real sample. Figure 10 shows the OM of the partial decapped sample, before and after XFIB-CE images on the targeted M3 location. The cross-sectional SEM image of cut M2 was shown in Figure 11, sample surface deposit a layer of insulator before testing. The testing result at different temperatures for the XFIB-CE sample was shown in Figure 12, the leakage current value of the test result was reduced from $\sim 2.5\text{mA}$ to μA range. The design team was very satisfied with the result, the results met their expectation and hypothesis. Multiple XFIB-CE samples have similar results, this technique shows a high success rate for case studies 1 and 2.

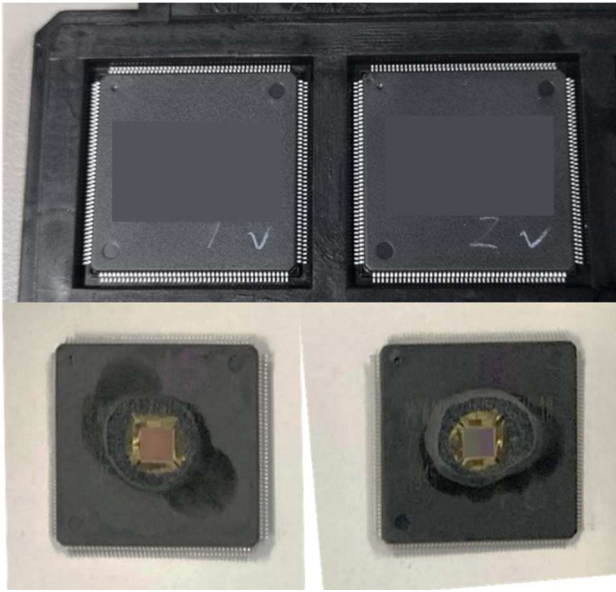


Fig. 9. Photo of the received lead-frame package sample, and after partial decapped samples for XFIB-CE and re-test after XFIB-CE.

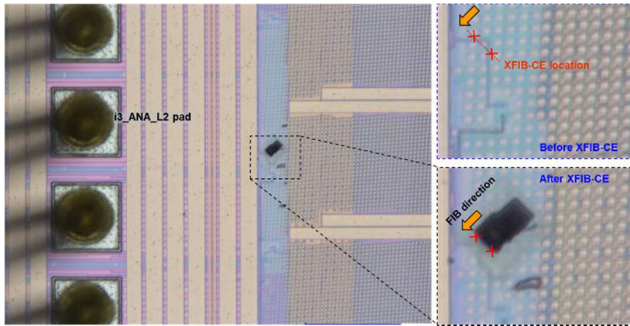


Fig. 10. OM images of the XFIB-CE targeted M3 location and after XFIB-CE image.

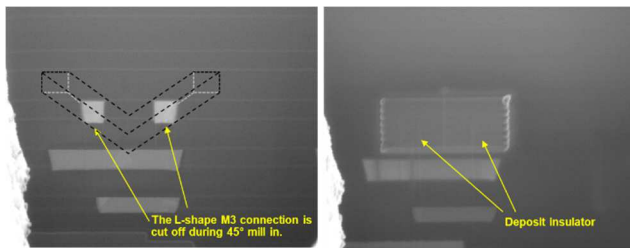


Fig. 11. Cross-sectional SEM image of the cut M3 (left), sample deposited insulator to prevent Cu oxidation (right).

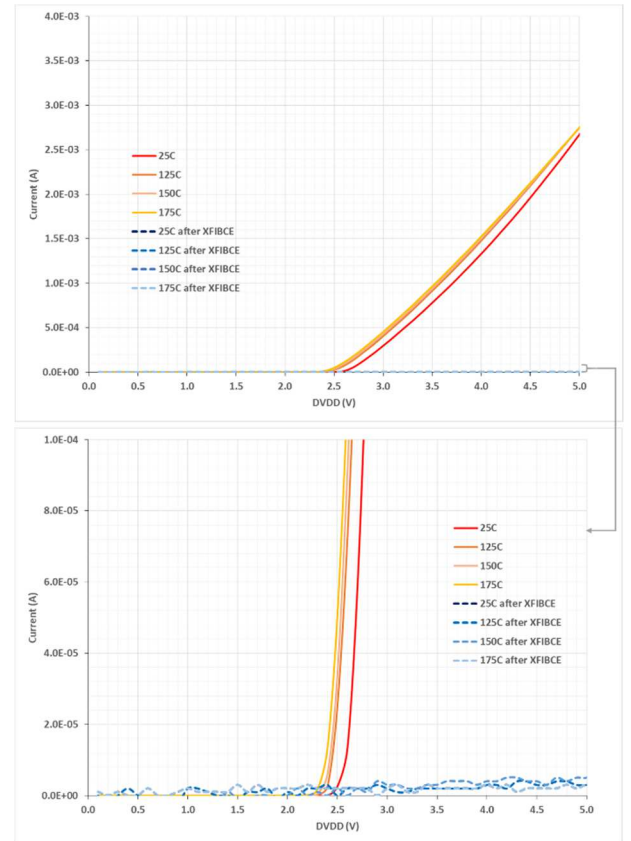


Fig. 12. Testing results on the XFIB-CE package at different temperatures. No leakage was found on the analog device.

C. Case study 3

Case study 3 is a customer return 40nm process test chip that failed electrostatic discharge (ESD) after Charged-Device Model (CDM) test [7]. To validate the design, the customer requested to FIB-CE remove multiple M3 connections to isolate other circuitry and test on the remaining standalone RF cells. Figure 13 shows the GDS layout of the FIB-CE location. This job is more challenging because the M3 is located near the wire bond pad, and the targeted M3 is below a thick M6 ($\sim 1\mu\text{m}$ thickness). Top-down FIB-CE is almost impossible for our FIB system because the GIS will hit the gold wire when inserting different types of GIS, such as insulator mill, Cu metal mill, and Pt deposition. Figure 14 shows an example of inserted GIS trigger touch alarm at the eccentric height in the system. Besides, it also causes gold wires to lift off or mechanical induced wire to cross short on some samples. To resolve this problem, we applied the XFIB-CE near the pad location. This surface is coated with polyimide and the depth of the M3 from the process surface (excluding the polyimide layer) is around $\sim 6.4\mu\text{m}$. Before XFIB-CE, the polyimide layer was removed with a reactive ion etcher (RIE). Figure 15 shows the cross-sectional image of the XFIB-CE recipe developed, the M3 was successfully isolated. In the real sample, to protect the gold wires from GIS, no Pt deposition was used. The sample was direct XFIB-CE at the interest location. To protect the cross-sectional surface, the insulator was deposited with an e-beam at an 0° stage angle, and the GIS is placed opposite the gold wire. Figure 16 shows the OM and SEM images of the XFIB-CE sample. The XFIB-CE samples were sent

back to the customer for testing. Customer feedback 2 out of 6 samples successfully passed the ESD CDM test, and 4 dies failed in the initial test before the CDM test. The 4 failed initial test chips were suspected got leakage induced by re-dep. The results help the customer to conclude no design issue, because got 2 chips passed the CDM test. This is another successful FIB-CE case.

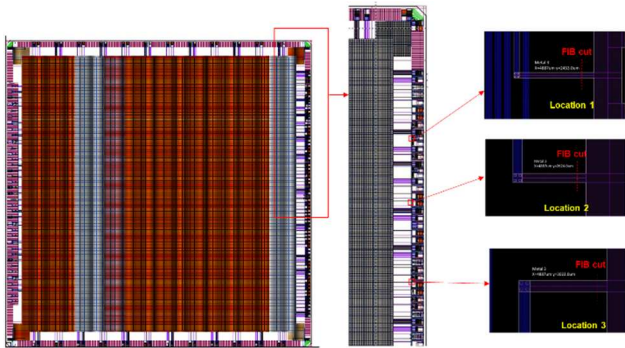


Fig. 13. Suggested FIB cut locations after studying the GDS layout, this is the biggest spacing and easiest isolation location available for the chip.

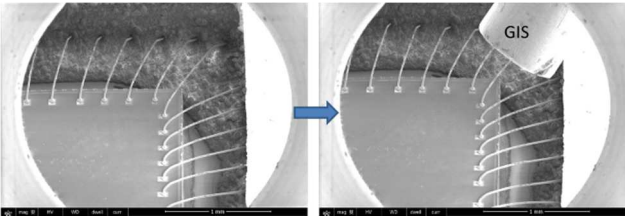


Fig. 14. An example of inserted GIS trigger touch alarm at the eccentric height in the system.

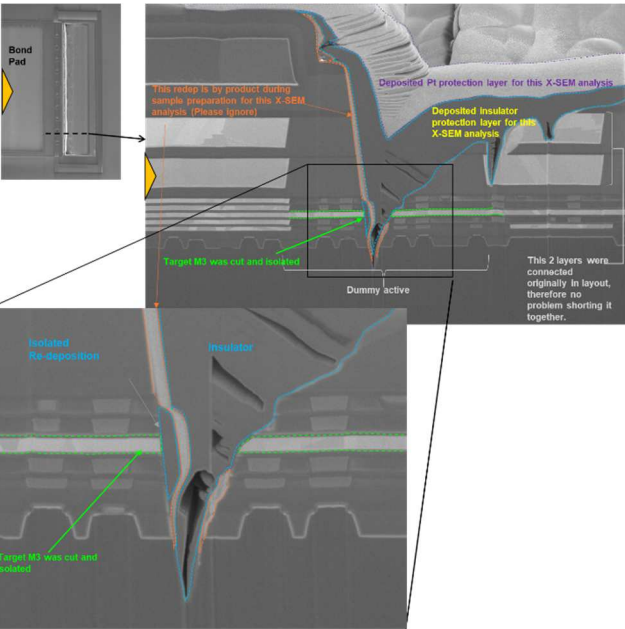


Fig. 15. XSEM images of the developed recipe of XFIB-CE cut, the targeted M3 was successfully cut and isolated.

IV. DISCUSSION

From all 3 case studies, there is no doubt that XFIB-CE is an alternate choice of FIB-CE techniques. It is simple and efficient as compared with top-down FIB-CE. However, it

has its limitation. XFIB-CE requires a clear spacing, an area without important metal line running except for dummy metals. Table 1 shows the comparison summary table of normal top-down FIB-CE and XFIB-CE using a normal standard dual-beam FIB system.

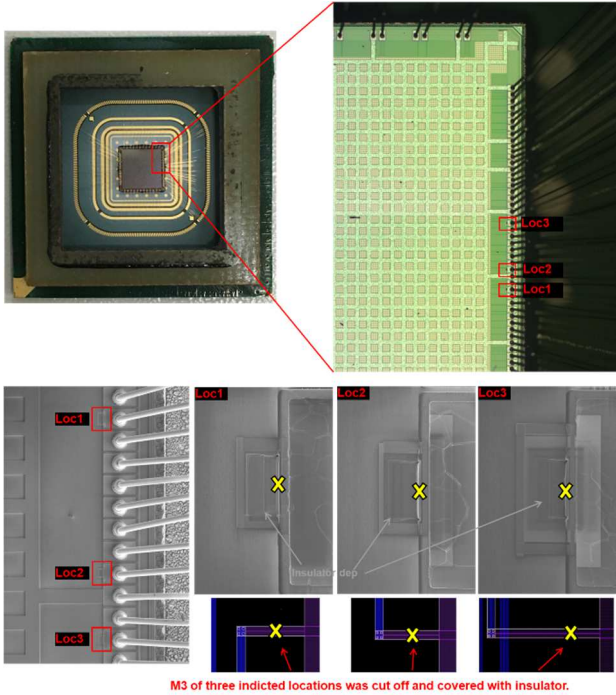


Fig. 16. OM and XSEM images of the XFIB-CE sample, 3 locations M3 was XFIB followed by insulator deposition by SEM e-beam.

Table 1. Comparison between XFIB-CE and top-down FIB-CE techniques.

FIB-CE with standard dual-beam FIB system		
	XFIB-CE	Top-down FIB-CE
Easiness/ Complexity. (1=Easiest, 2, 3, 4, 5=Hardest)	Normal cross-sectional, 2	4
Require etch GIS (Metal etch, Insulator etch)	No	Yes
Efficiency. (1=Most efficient, 2, 3, 4, 5=Less efficient)	1	Case by case, 3
Wafer and package level FIB	Yes	Yes
Easiness to XFIB thick metal $\geq 3\mu\text{m}$. (1=Easiest, 2, 3, 4, 5=Hardest)	2	4
Cut metal surrounding requirement.	No important circuitry metals beside targeted metal	Case by case

V. CONCLUSION

Through the three case studies, XFIB-CE is proven a simple and efficient alternative circuit isolation technique, especially when a batch of samples is required. The novel XFIB-CE technique also demonstrated workability on the wafer and package level. The XFIB-CE technique is a useful technique for a company without an expensive dedicated specialized FIB-CE system, only using a standard dual-beam FIB system. In other words, more economical.

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